

Cabela, Razeel V.
BSIT - 4E

Enterprise Infrastructure Plan
ABC Startup Solutions

Company Name: ABC Startup Solutions

Nature of Business

ABC Startup Solutions is a startup software development company that provides web development, mobile application development, custom business software, and IT solutions for small and medium-sized businesses.

Company Vision

To become a trusted and innovative software development company that provides reliable, secure, and efficient digital solutions to businesses in the Philippines.

Company Mission

ABC Startup Solutions aims to develop high-quality software solutions, provide excellent customer service, use modern technologies, protect company and client information, and maintain a secure and efficient IT environment for its employees and customers.

Office Location

Unit 201, Startup Business Center, Santa Cruz, Laguna, Philippines

Company Management

- Information Technology Department – 5 Employees
- Human Resources Department – 4 Employees
- Finance Department – 5 Employees
- Sales Department – 6 Employees

Department	Suggested Employees
IT	1 System Administrator, 1 Helpdesk Technician, 3 Software Developers
HR	1 HR Officer, 3 HR Staff
Finance	1 Finance Manager, 3 Accountants, 1 Finance Assistant
Sales	1 Sales Manager, 5 Sales Representatives

ENTERPRISE HARDWARE INVENTORY

Asset ID	Hardware	Qty.	Department/Area	Purpose
HW-001	Desktop Computers	16	IT, HR, Finance, Sales	Primary employee workstations
HW-002	Business Laptops	4	IT and Sales	Mobile work, meetings and remote access
HW-003	Server	1	IT / Server Area	Centralized file, application and network services
HW-004	Router	1	IT / Network Area	Routes traffic between the company network and internet
HW-005	48-Port Managed Gigabit Switch	1	IT / Network Area	Connects wired computers and network devices
HW-006	Multifunction Network Printer	2	Office	Printing, scanning and copying documents
HW-007	1500VA UPS	2	Server/Network Area	Backup electricity and protection for critical equipment
HW-008	Wireless Access Point	2	Office	Provides wireless network coverage
HW-009	NAS Storage	1	IT / Server Area	Centralized backup and additional file storage
HW-010	External Backup Drive 4 TB	2	IT	Offline backup copies
HW-011	24-inch LED Monitors	20	All Departments	Employee display devices
HW-012	Keyboard and Mouse Sets	20	All Departments	Workstation input devices
HW-013	Server Rack	1	IT / Server Area	Organizes server and networking equipment

ABC Startup Solutions requires 20 employee computers because the company has 20 employees. Sixteen desktop computers will be used for employees who primarily work inside the office, while four laptops will provide mobility for IT personnel and sales employees. A centralized server and NAS will support company files, services, and backups. A 48-port managed switch provides enough network ports for the current infrastructure while leaving room for future expansion. Two wireless access points will provide sufficient Wi-Fi coverage throughout the single office floor. UPS units are included to protect the server and networking equipment during power interruptions.

ENTERPRISE SOFTWARE INVENTORY

Software	Version	License	Purpose
Windows 11 Pro	Current Supported Release	Commercial	Operating system for employee computers
Ubuntu Server	LTS Release	Open Source	Server operating system
Microsoft Office	Microsoft 365 / Current Release	Commercial	Documents, spreadsheets and presentations
Visual Studio Code	Current Stable Release	Free	Software development and programming
Git	Current Stable Release	Open Source	Source-code version control
GitHub Desktop	Current Stable Release	Free	Graphical Git and GitHub management
VirtualBox	Current Stable Release	Open Source	Creates virtual machines for testing
Google Chrome	Current Stable Release	Free	Web browsing and web application testing
Microsoft Defender	Current Release	Included/Commercial	Malware and endpoint protection
AnyDesk	Current Release	Commercial Business License	Remote technical support
7-Zip	Current Stable Release	Open Source	File compression and extraction

Windows 11 Pro will serve as the primary workstation operating system because it provides business and security features. Ubuntu Server will be used for server-based services because of its stability and support for development tools. Microsoft Office is required for common office productivity tasks. VS Code, Git and GitHub Desktop are important for software development and version control. VirtualBox allows the IT team to test operating systems and applications in virtual environments. Google Chrome supports browsing and web application testing, while Microsoft Defender provides endpoint security. AnyDesk will allow authorized IT personnel to provide remote support, and 7-Zip will be used for file compression and extraction.

ENTERPRISE NETWORK INVENTORY

Network Equipment	Quantity	Purpose
ISP Fiber Modem/ONT	1	Provides connection from the ISP
Business Router	1	Routes internet and internal network traffic
Hardware Firewall	1	Protects the internal network from unauthorized traffic
48-Port Managed Gigabit Switch	1	Connects wired network devices
Wireless Access Point	2	Provides office Wi-Fi
24-Port CAT6 Patch Panel	2	Organizes up to 48 network connections
CAT6 UTP Cable	1 × 305-meter box	
RJ45 Connectors	100 pieces	Terminates Ethernet cables
CAT6 Patch Cords	30	Connects network equipment and wall ports
Network Rack	1	Houses network equipment
Cable Management Panel	2	Organizes network cables

A 48-port network design is recommended because the company currently has 20 employees but also needs connections for servers, printers, wireless access points and other network devices. The additional available ports will support future expansion without requiring the company to immediately replace the main switch. CAT6 cabling provides reliable Gigabit Ethernet connectivity throughout the office.

SYSTEM ADMINISTRATION ROLES

A. Helpdesk Technician

Responsibilities

- Provides first-level technical support to employees.
- Troubleshoots computer, software, printer and basic network problems.
- Installs and configures applications.
- Creates and manages support tickets.
- Assists users with account and password problems.
- Escalates complicated problems to administrators.

Skills

- Hardware and software troubleshooting
- Windows administration
- Basic networking
- Customer service
- Communication
- Problem solving

Common Tools

- AnyDesk
- Remote Desktop
- Helpdesk ticketing software
- Windows Event Viewer
- Command Prompt
- PowerShell

Certifications

- CompTIA A+
- CompTIA Network+
- Microsoft technical certifications

B. Network Administrator

Responsibilities

- Installs and manages routers, switches and wireless access points.
- Maintains LAN and internet connectivity.
- Configures network security.
- Troubleshoots network problems.
- Monitors network performance.
- Manages IP addresses, VLANs and network devices.

Skills

- TCP/IP
- Subnetting
- Routing and switching
- VLAN configuration
- Firewall administration

- Network troubleshooting

Common Tools

- Cisco IOS
- PuTTY
- Wireshark
- Ping
- Traceroute
- Nmap
- Network monitoring software

Certifications

- Cisco Certified Network Associate (CCNA)
- CompTIA Network+
- CompTIA Security+

c. **Linux System Administrator**

Responsibilities

- Installs and maintains Linux servers.
- Creates and manages user accounts.
- Configures server applications and services.
- Performs system updates.
- Monitors server performance.
- Performs backups.
- Troubleshoots Linux problems.
- Manages file permissions and system security.

Skills

- Linux command line
- Bash scripting
- User and permission management
- Networking
- Server administration
- Troubleshooting
- Security

Common Tools

- Ubuntu Server
- SSH
- Bash
- systemctl
- Git
- Vim or Nano
- Cron
- rsync

Certifications

- Red Hat Certified System Administrator (RHCSA)

- Linux Foundation Certified System Administrator (LFCS)
- CompTIA Linux+

D. Cloud Administrator

Responsibilities

- Manages cloud computing resources.
- Creates and manages cloud servers.
- Manages cloud storage.
- Configures user access and permissions.
- Monitors cloud services.
- Creates backups and recovery procedures.
- Maintains cloud security.
- Controls cloud infrastructure costs.

Skills

- Cloud computing
- Virtualization
- Networking
- Identity and access management
- Security
- Backup and disaster recovery
- Monitoring

Common Tools

- Microsoft Azure
- Amazon Web Services
- Google Cloud
- PowerShell
- Cloud monitoring tools
- Virtual machines

Certifications

- Microsoft Azure Administrator
- AWS Certified SysOps Administrator
- Google Cloud Associate Cloud Engineer

How the Four IT Professionals Work Together

The Helpdesk Technician, Network Administrator, Linux System Administrator and Cloud Administrator work together to maintain the organization's IT services. The Helpdesk Technician serves as the first point of contact when employees experience technical problems. Network-related problems are escalated to the Network Administrator, who maintains routers, switches, wireless connections and network security. The Linux System Administrator manages Linux servers, user permissions, services and server security. The Cloud Administrator manages cloud-based servers, applications, storage and backup services. By sharing information and coordinating their responsibilities, these professionals help maintain a reliable, secure and efficient IT infrastructure for the organization.

INFRASTRUCTURE RECOMMENDATIONS

Internet Provider

Recommended Provider: PLDT Enterprise Fiber

Recommended Connection: Business fiber internet with at least 500 Mbps download/upload capacity

A business-grade fiber connection is recommended because ABC Startup Solutions depends heavily on internet access for software development, cloud services, video meetings, email, GitHub and communication with clients. A secondary internet connection from another provider should also be considered as a backup to maintain connectivity if the primary ISP experiences an outage.

Server Specifications:

Component	Recommended Specification
Processor	Intel Xeon or equivalent server-grade 8-core processor
RAM	64 GB ECC RAM
Operating System	Ubuntu Server LTS
OS Storage	2 × 1 TB SSD configured as RAID 1
Data Storage	2 × 4 TB enterprise HDD/SSD configured as RAID 1
Network	Dual Gigabit Ethernet
Power Supply	Redundant server-grade PSU
Backup Power	1500VA or higher UPS

The recommended server specifications provide enough processing power and memory for the company's initial 20 users while allowing additional services and virtual machines to be deployed later. RAID 1 provides disk redundancy so that the system can continue operating if one drive fails.

Backup Strategy:

Use the **3-2-1 Backup Strategy**:

Original Company Data



Local NAS Backup



External Backup Drive



Encrypted Off-site / Cloud Backup

The company should follow the **3-2-1 backup strategy** by maintaining at least three copies of important data, storing the copies on at least two different types of storage media, and keeping one backup copy off-site. Automatic incremental backups should be performed daily to the NAS, while a full backup should be performed weekly. An encrypted external or cloud backup should be maintained to protect the company against hardware failure, theft, malware and other disasters.

Security Recommendations:

- Hardware firewall
- VLAN separation between departments
- Multi-factor authentication
- Regular software updates
- User access control
- Least-privilege access
- Automatic screen locking
- Data encryption
- Employee cybersecurity awareness training
- Regular backups
- Network monitoring
- Disable unused accounts
- Secure Wi-Fi using WPA2/WPA3

Security should be implemented using multiple layers rather than depending on a single security tool. The firewall will control network traffic, while access controls, VLANs and authentication will prevent users from accessing resources they do not need. Regular updates, backups and security awareness training will further reduce the risk of malware, unauthorized access and data loss.

Antivirus:

Recommended Antivirus/Endpoint Protection: Microsoft Defender

Microsoft Defender should be enabled on all Windows workstations to detect malware, ransomware and other threats. Antivirus signatures and security updates should be installed automatically. The IT department should regularly check security alerts and ensure that protection remains enabled on all computers.

Password Policy:

ABC Startup Solutions Password Policy

- Minimum password length: 12 characters
- Must use a combination of letters, numbers and special characters.
- Employees should not share passwords.
- Password reuse should not be allowed for important accounts.
- Multi-factor authentication should be enabled for email, GitHub, administrative and cloud accounts.
- Default passwords must immediately be changed.

- Administrative accounts should be separate from normal user accounts.
- Passwords should be stored using an approved password manager.
- Accounts should be temporarily locked after repeated unsuccessful login attempts.

Expansion Plan:

The infrastructure should be designed to support company growth from 20 employees to approximately 40–50 employees. The 48-port managed switch provides additional network ports for future computers and devices. Additional switches and wireless access points can be installed if the company expands to another office area. Server memory and storage should be upgradeable, and virtualization can be used to deploy additional servers without purchasing separate physical machines for every service. Cloud services may also be introduced as the organization grows. Software licensing, backup capacity and internet bandwidth should be reviewed regularly to ensure that they remain sufficient for the increasing number of employees.

Personal Reflection

Through this Enterprise Infrastructure Planning project, I learned that building an IT infrastructure requires more than simply purchasing computers and connecting them to the internet. A System Administrator must first understand the needs of the organization, the number of employees, the departments and the services that the company will use. I learned how hardware, software, servers, networking devices, security and backup systems are connected to each other and how they support the daily operations of a company.

The most challenging part of the activity for me was designing the enterprise network diagram. I had to understand where the ISP modem, router, firewall, switch, wireless access points, server, printer and employee computers should be placed in the network. I also learned that network devices must be connected in a logical order. Creating the diagram helped me understand how internet traffic reaches the company's devices and how different departments can be connected through a centralized network.

I also realized why infrastructure planning is important before deployment. Without proper planning, a company may purchase unnecessary equipment, select hardware that cannot support future expansion, create security vulnerabilities or experience network problems. Planning allows the System Administrator to estimate the required equipment, prepare an appropriate network design, establish backup procedures and implement security policies before the infrastructure is used by employees.

This project also helped me understand the different responsibilities of IT professionals. Helpdesk Technicians assist users with common technical problems, Network Administrators maintain network connectivity, Linux System Administrators manage servers and Cloud Administrators maintain cloud resources. Although they have different responsibilities, they must work together to maintain reliable IT services.

Overall, this project improved my understanding of system administration and infrastructure planning. It allowed me to apply concepts related to hardware, software, networking, security, backups and technical documentation. These skills will help me become a better System Administrator because I now have a clearer understanding of how to plan an IT environment before installing and configuring actual equipment.

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